

Continue and describe sequences

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Continue 7, 12, 17, 22 and state the rule. Copy or complete the first step.

2. Find the missing terms: 84, __, 62, __, 40.

3. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Continue 7, 12, 17, 22 and state the rule. Copy or complete the first step.
Answer 27, 32; add 5.
2. Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51.
3. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer $C = 6n + 4$.
4. Miro says: Miro treats a variable as a label instead of a number that can vary. Is this correct? Explain.
Answer State what the variable represents, substitute its value and preserve the operation structure.

Continue and describe sequences

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Continue 7, 12, 17, 22 and state the rule.

6. Check this result using an inverse, estimate or known fact:
Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

2. Find the missing terms: 84, __, 62, __, 40.

7. Prove or explain your reasoning: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

3. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.

4. Solve this independently and show every stage: Continue 7, 12, 17, 22 and state the rule.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the missing terms: 84, __, 62, __, 40.

10. Write and solve a similar question to: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

Teacher answers and checking prompts

1. Continue 7, 12, 17, 22 and state the rule.
Answer 27, 32; add 5.
2. Find the missing terms: 84, __, 62, __, 40.
Answer 73 and 51.
3. Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer $C = 6n + 4$.
4. Solve this independently and show every stage: Continue 7, 12, 17, 22 and state the rule.
Answer 27, 32; add 5.
5. Use a second method or representation for: Find the missing terms: 84, __, 62, __, 40.
Answer Expected result: 73 and 51. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer Expected result: $C = 6n + 4$. A valid check must be shown.
7. Prove or explain your reasoning: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31. The explanation must show why the method works.
8. Identify and correct this misconception: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Continue and describe sequences

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.

6. Create a new example that tests the same idea as: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.

2. Prove the answer to this question, rather than only calculating it: Continue 7, 12, 17, 22 and state the rule.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the missing terms: 84, __, 62, __, 40.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $C = 6n + 4$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find A when $A = 3b + 2c$, $b = 7$ and $c = 5$.
Answer 31. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Continue 7, 12, 17, 22 and state the rule.
Answer Expected result: 27, 32; add 5. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the missing terms: 84, __, 62, __, 40.
Answer Expected result: 73 and 51. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $C = 6n + 4$.
Answer One valid reconstruction is: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro treats a variable as a label instead of a number that can vary.
Answer State what the variable represents, substitute its value and preserve the operation structure.
6. Create a new example that tests the same idea as: Write a formula for the cost C of n tickets at £6 each plus a £4 booking fee.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.